

4.9 Newton's Method Worksheet

1. Approximate $\sqrt{6}$ to four decimal places.

Let $f(x) = x^2 - 6$ therefore $f(\sqrt{6}) = 0$ and $f'(x) = 2x$. We make our initial guess based on the fact that $\sqrt{4} = 2$. So we guess that $x_1 = 2.5$.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.4500$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \approx 2.4495$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} \approx 2.4495$$

2. Approximate e^2 to four decimal places.

Let $f(x) = \ln(x) - 2$ therefore $f(e^2) = 0$ and $f'(x) = \frac{1}{x}$. We make our initial guess based on the fact that $e \approx 2.72$. So we guess that $x_1 = 6.5$.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \approx 7.3333$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \approx 7.3888$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} \approx 7.3891$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} \approx 7.3891$$

3. Approximate $\sqrt[\pi]{e}$ to five decimal places.

Let $f(x) = x^\pi - e$ therefore $f(\sqrt[\pi]{e}) = 0$ and $f'(x) = \pi x^{\pi-1}$. We make our initial guess based on the fact that $e \approx 2.72$ and $\pi \approx 3.14$. So we guess that $x_1 = 1.5$.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \approx 1.38564$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \approx 1.37489$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} \approx 1.37480$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} \approx 1.37480$$

4. Find all solutions of the equation $\frac{1}{x} = 1 + x^3$ (up to six decimal places).

By graphing these functions we see that they only have one intersection.

Let $f(x) = \frac{1}{x} - 1 - x^3$. Then $f'(x) = -\frac{1}{x^2} - 3x^2$.

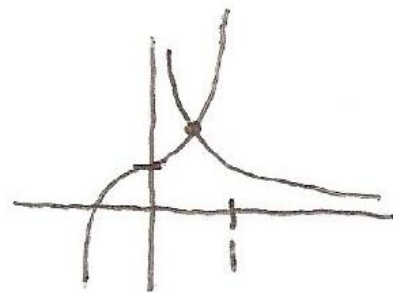
We make our initial guess less than 1. Suppose we choose $x_1 = 0.5$.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \approx 0.684211$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \approx 0.724100$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} \approx 0.724492$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} \approx 0.724492$$



5. Consider the function $f(x) = x^3 - x - 3$.

- a) Show that Newton's Method does not find a root if $x_1 = 0$. What does the sequence of approximation do here? Round decimals to six places.

$$x_1 = 0$$

$$x_2 = -3$$

$$x_3 \approx -1.961538$$

$$x_4 \approx -1.147175$$

$$x_5 \approx -0.006579$$

$$x_6 \approx -3.000389$$

$$x_7 \approx -1.961818$$

$$x_8 \approx -1.147430$$

$$x_9 \approx -0.007256$$

$$x_{10} \approx -3.000473$$

If we look carefully we can see that this sequence of approximations appears to be looping.

- b) Show that Newton's Method can find a root if $x_1 = 2$. Find the root accurate to six places.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \approx 1.727272$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \approx 1.673691$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} \approx 1.671702$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} \approx 1.671699$$

$$x_6 = x_5 - \frac{f(x_5)}{f'(x_5)} \approx 1.671699$$

- c) What do parts (a) and (b) tell you about the selection of x_1 ?

This tells us that our guess for x_1 is important and we must try to have an initial value that is close to the root we are trying to approximate.